AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT101
General Mathematics

Midterm Examination (MTA) - Makeup
Spring Semester 2024/2025

Date: 23 / April /2025

Number of Exam Pages:2 (including this cover sheet)	**Time Allowed: 90 minutes**

Instructions:

- Total Marks:**60 marks**
- This exam consists of **4 questions**
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Scientific Calculators** are allowed.

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ESSAY QUESTIONS

Each question is worth 15 marks. You should answer all the questions, giving 60 possible marks.

Q–1: [4+5+6 Marks]

a) let f(x)= $\sqrt{x}$, and h(x)= x-1

i) Evaluate (f – h) (9)

ii) Evaluate (f o h) (x) and find (f o g)(2).

Solve for x: 2 (5+x)- 7 = 3(x-2)

Q–2: [8+7Marks] solve the following:

a) 5 | 2x-3 |< 10 (write the solution in interval notation)

b) $-4 \leq \frac{2x-3}{3} < 4$ (write the solution in interval notation)

Q–3: [6+7+2] Marks]

a) solve the following equations:

i) $\sqrt[3]{2x-5} - 4 = -1$

ii) $2x^2 + 5x + 3 = 0$ (use the quadratic formula)

b) Find the domain of f(x)= $\sqrt{3x-4}$

Q–4: [8+7Marks]

a) Write the equation of a circle in standard form, and identify the center and the radius. $2x^2 + 2y^2 - 2x - 2y - 3 = 0$

b) Find the equation of the line that passes through the two points A(3, -2),and B(-3, -2).

End of the questions.

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